Ingénierie des circuits intégrés

MASTER IESE

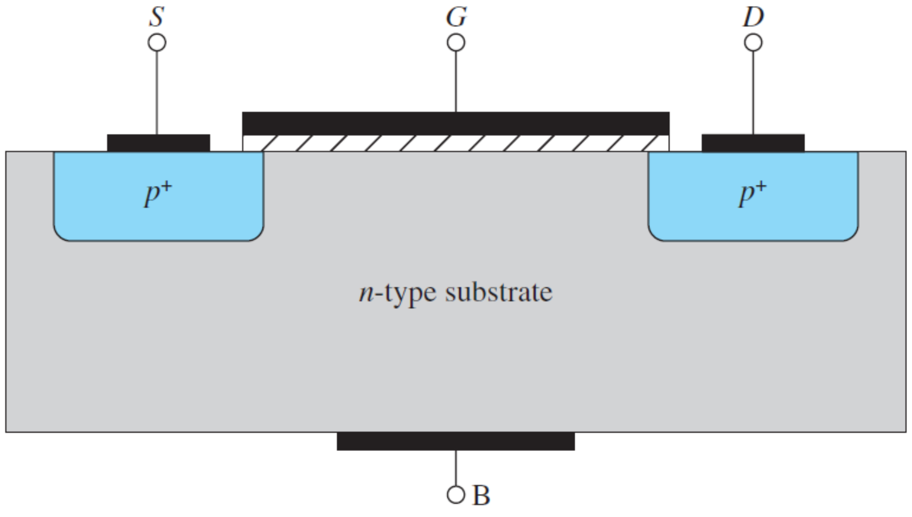
2025-2026

Partie 3

- Technology CMOS
- Transistor MOSFET operation, NMOS and PMOS
- Application, NMOS Transistor as amplifier
- Linear amplifier in technology CMOS
- Base Integrated circuits
  - ✓ Transistor in diode connected structure
  - ✓ Current source
  - ✓ Current Mirror

p-Channel MOSFET

- Changing the doping polarities of the substrate and the S/D areas results in a “PMOS” device



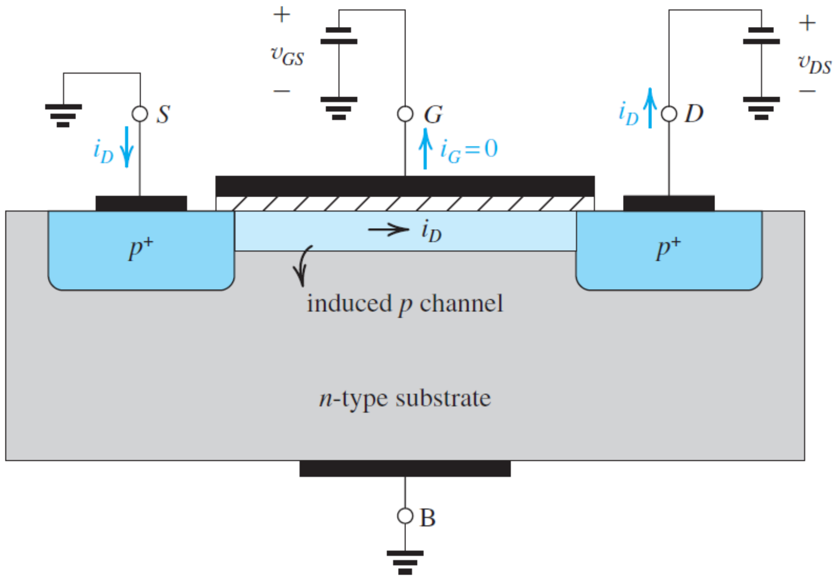
- Figure at right shows a cross-sectional view of a *p*-channel enhancement-type MOSFET.
- The structure is similar to that of the NMOS device except that here the substrate is *n* type and the source and the drain regions are *p*+ type;
- The PMOS and NMOS transistors are said to be *complementary* devices.

p-Channel MOSFET

- All semiconductor regions are reversed in polarity relative to their counterparts in the NMOS case.
- A negative voltage v_{GS} of magnitude greater than $|V_{tp}|$ induces a *P* channel, and a negative v_{DS} causes a current i_D to flow from source to drain.

- To induce a channel for current flow between source and drain, a negative voltage is applied to the gate, that is, between gate and source
- By increasing the magnitude of the negative v_{GS} beyond the magnitude of the threshold voltage Vtp, which by convention is negative, a p channel is established as shown in Fig. at right.
- This condition can be described as

$$v_{GS} \leq V_{tp} \qquad |v_{GS}| \geq |V_{tp}|$$



p-Channel MOSFET

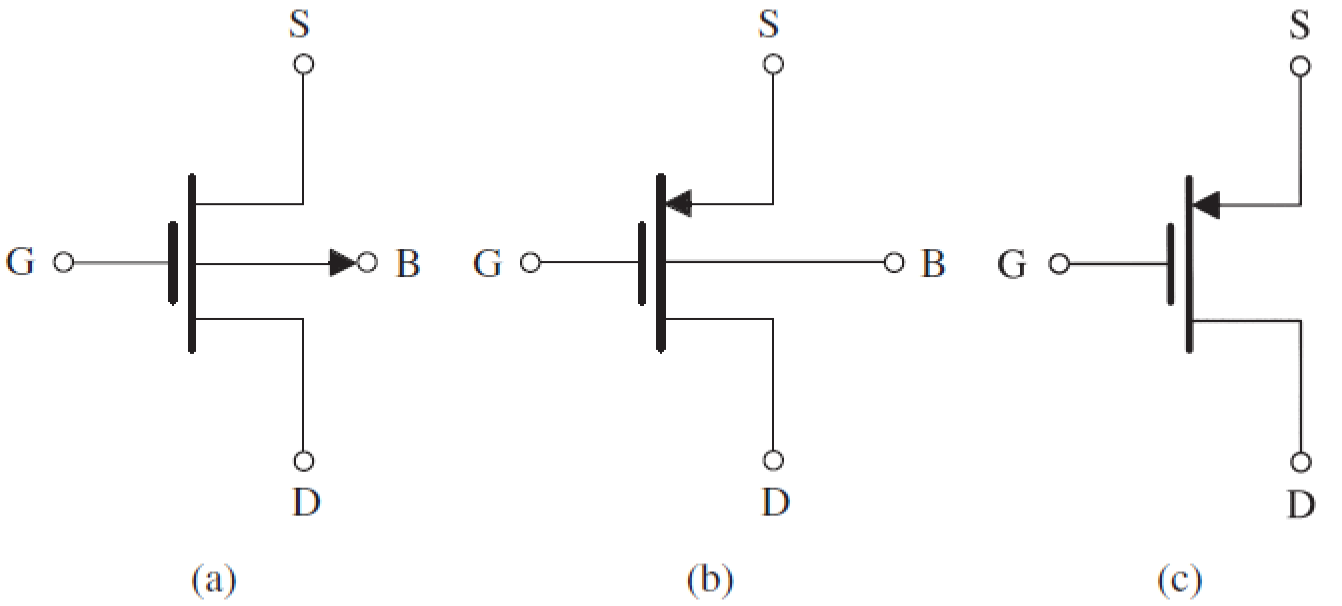
- To cause a current i_D to flow in the p channel, a negative voltage v_{DS} is applied to the drain.
- The current i_D is carried by holes and flows through the channel from source to drain.
- As we have done for the NMOS transistor, we define the process transconductance parameter for the PMOS device as $k'_p = \mu_p C_{ox}$
- where μ_p is the mobility of the holes in the induced p channel. Typically, $\mu_p = 0.25*\mu_n$ to $0.5\mu_n$ and is process-technology dependent.
- The transistor transconductance parameter k_p is obtained by multiplying k'_p by the aspect ratio W/L,

$$k_p = k'_p(W/L)$$

Characteristics of the p-Channel MOSFET

- The circuit symbol for the p-channel enhancement-type MOSFET is shown in Fig
- in the saturation-region expression for i_D as follows

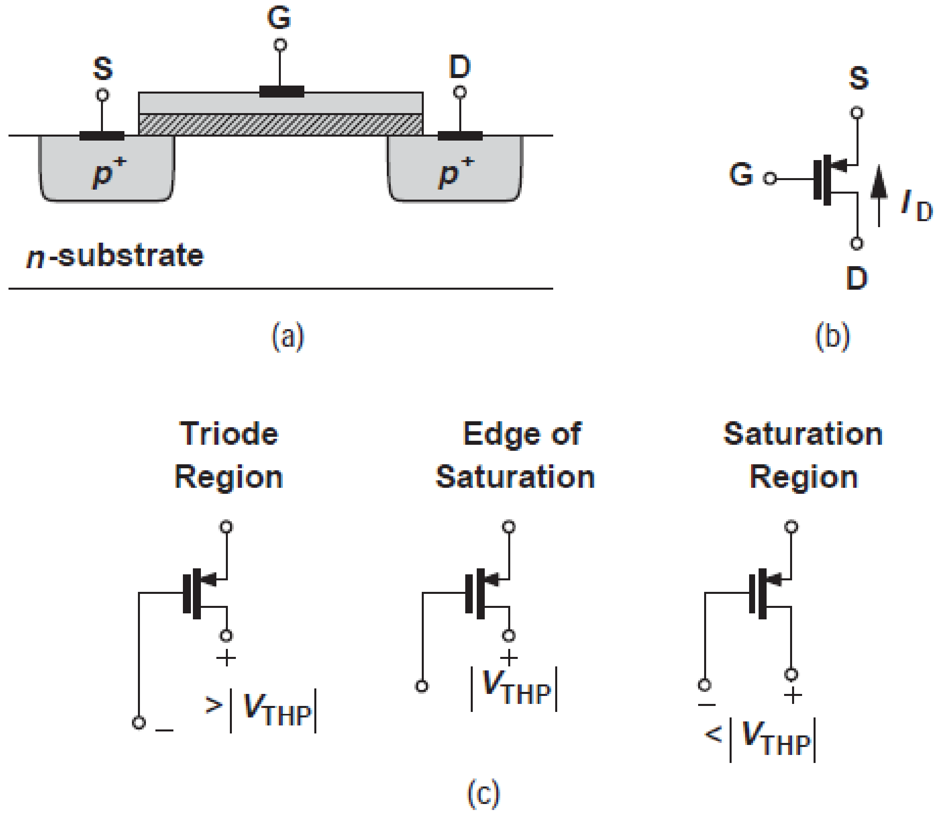
$$i_D = \frac{1}{2} k_p' \left(\frac{W}{L}\right) \left(v_{SG} - |V_{tp}|\right)^2$$



(a) Circuit symbol for the *p*-channel enhancement-type MOSFET. **(b)** Modified symbol with an arrowhead on the source lead. **(c)** Simplified circuit symbol for the case where the source is connected to the body.

Characteristics of the p-Channel MOSFET

- The circuit symbol for the p-channel enhancement-type MOSFET

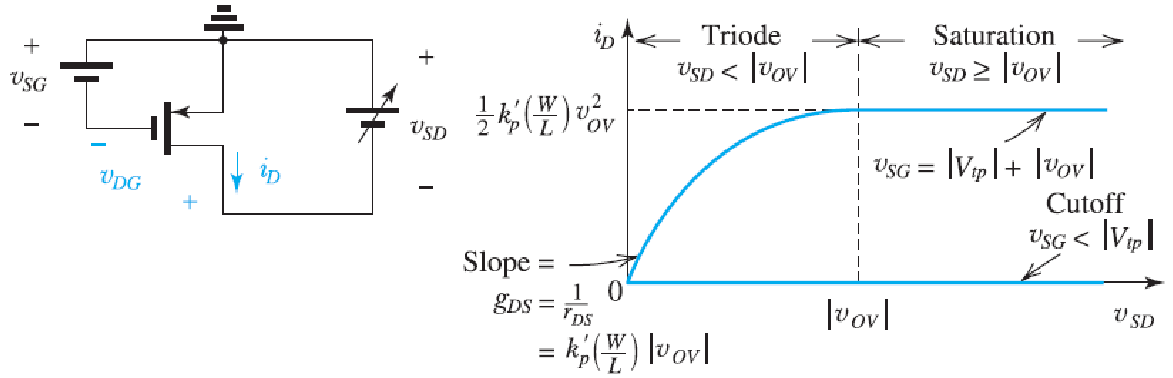


- Changing the doping polarities of the substrate and the S/D areas results in a “PMOS” device fig (a).
- The channel consists of *holes* and is formed if the gate voltage is *below* the source potential by one threshold voltage.
- That is, to turn the device on, $V_{GS} < V_{TH}$, where V_{TH} itself is negative.

(a) Structure of PMOS device, (b) PMOS circuit symbol, (c) illustration of triode and saturation regions based on gate and drain voltages.

Characteristics of the p-Channel MOSFET

Table 5.2 Regions of Operation of the Enhancement PMOS Transistor



- $v_{SG} < |V_{tp}|$: no channel; transistor in cutoff; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently

$$v_{SD} < |v_{OV}|$$

Then

$$i_D = k_p'\left(\frac{W}{L}\right)\left[(v_{SG} - |V_{tp}|)v_{SD} - \frac{1}{2}v_{SD}^2\right]$$

or equivalently

$$i_D = k_p'\left(\frac{W}{L}\right)\left(|v_{OV}| - \frac{1}{2}v_{SD}\right)v_{SD}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

Then

$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(v_{SG} - |V_{tp}|)^2$$

or equivalently

$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)v_{OV}^2$$

Finite Output Resistance in Saturation Region for p-channel MOSFET-

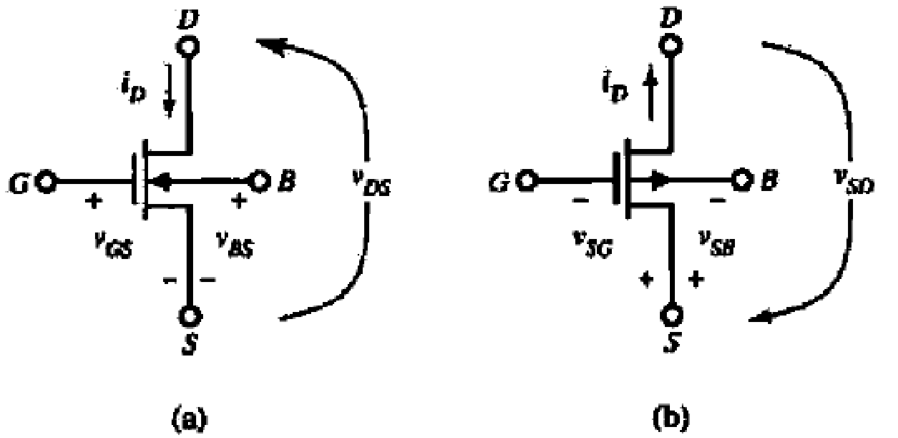
With channel length modulation, in the saturation region the current expressed as

$$i_D = \frac{1}{2} k_p' \left(\frac{W}{L}\right)\left(v_{SG} - |V_{tp}|\right)^2 (1 + |\lambda| v_{SD})$$

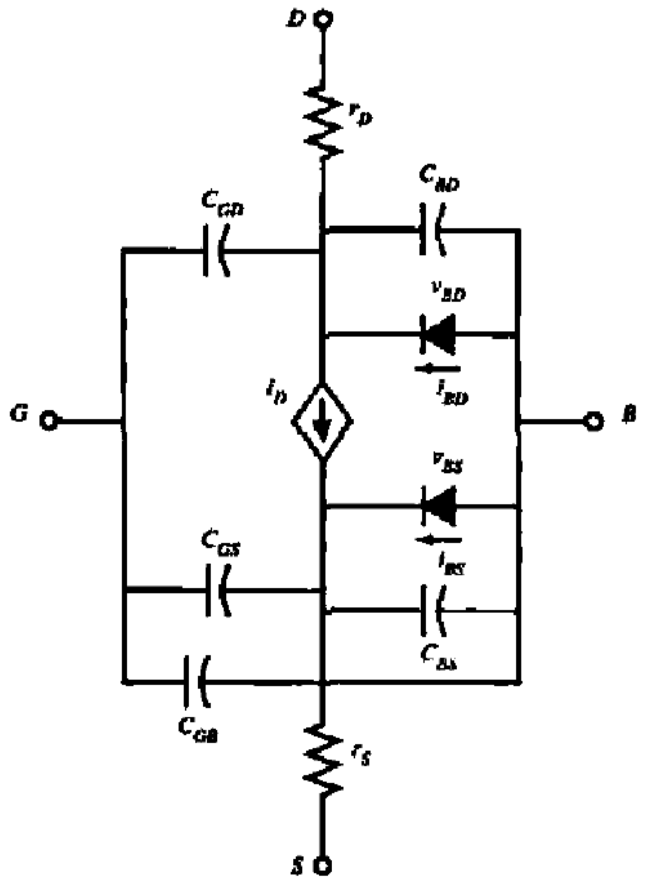
And the finite resistance is expressed as

$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1}_{v_{GS}\ \text{constant}} = \left[\lambda \frac{k_n'}{2} \frac{W}{L} (V_{GS} - V_{tn})^2\right]^{-1} = \frac{1}{\lambda I_D} = \frac{V_A}{I_D}$$

Symbol and Large signal Model for transistor MOSFET

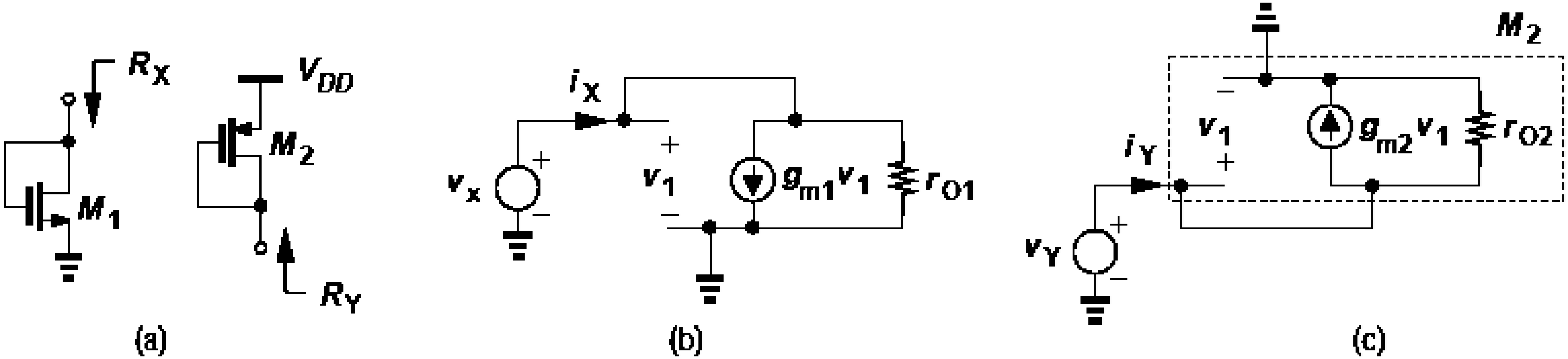


Positive convention for a) n-chanel and b) p-chanel MOS transistor



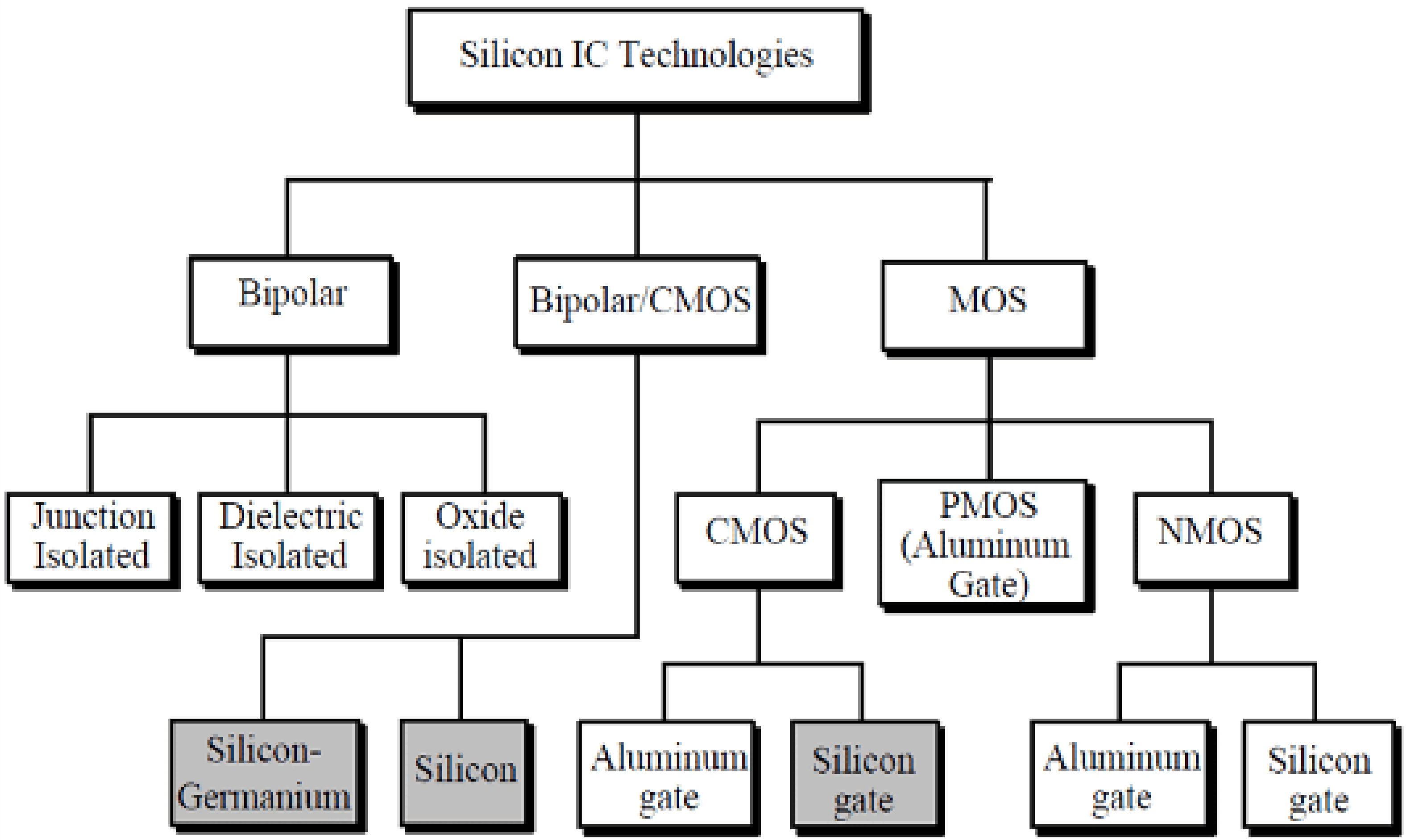
Complete large signal model for the MOS transistor

Small signal Model for transistor MOSFET



(a) Diode-connected NMOS and PMOS devices, (b) small-signal model of (a) n-channel, (c) small-signal model of (a) p-channel.

Categories of silicon technology

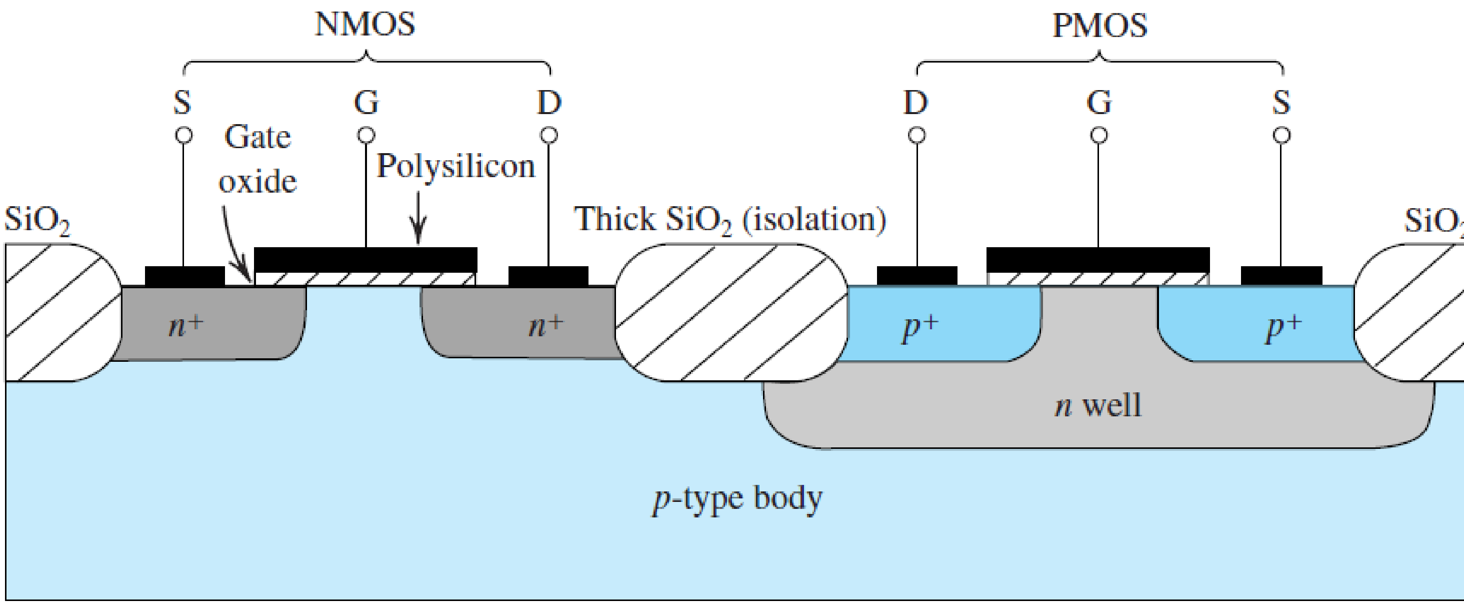


Complementary MOS or CMOS

- As the name implies, complementary MOS technology employs MOS transistors of both polarities. Although CMOS circuits are somewhat more difficult to fabricate than NMOS, the availability of complementary devices makes possible many powerful circuit configurations.

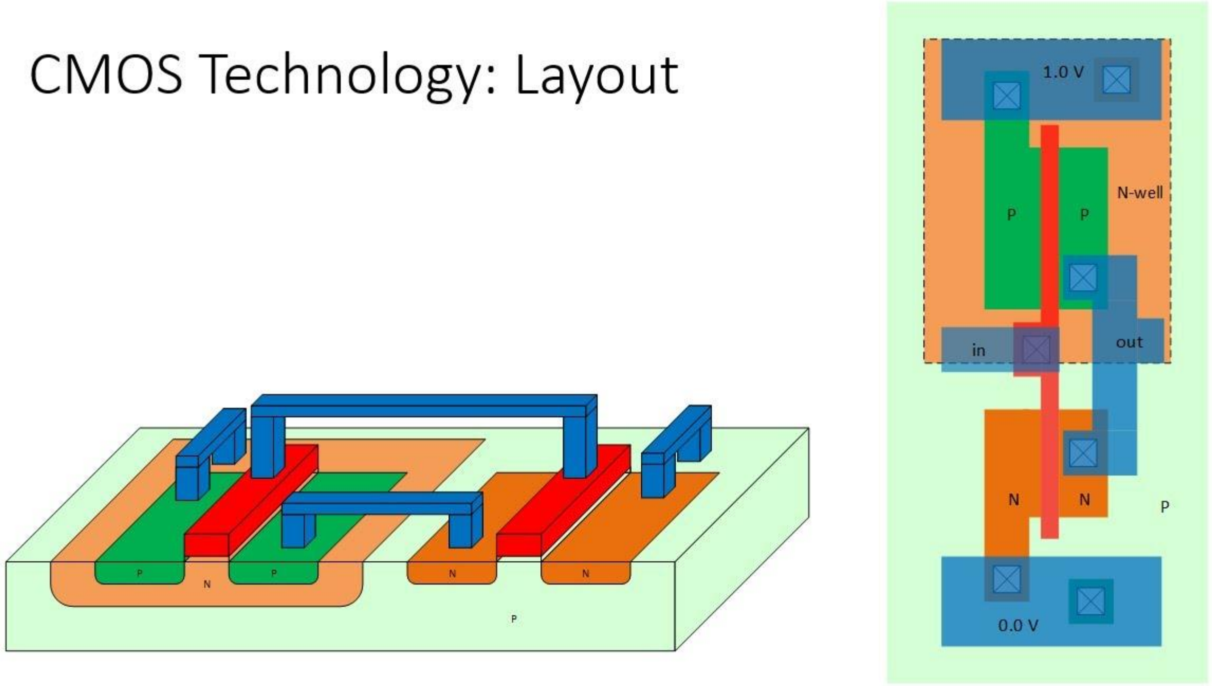
- At the present time CMOS is the most widely used of all the IC technologies and that for both analog and digital circuits.

- CMOS technology has virtually replaced designs based on NMOS transistors alone.



Cross section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate *n*-type region, known as an *n* well. Another arrangement is also possible in which an *n*-type substrate (body) is used and the *n* device is formed in a *p* well.

Complementary MOS or CMOS



NMOS and PMOS Transistors in CMOS Technology and Layout

CMOS Technology Applications

The MOS transistors are the **fundamental building blocks of modern integrated circuits (ICs)** — both analog and digital.

1. Digital Logic Circuits

MOS transistors form the basis of **CMOS (Complementary MOS)** technology, which dominates digital IC design.

- **Logic gates:** NAND, NOR, AND, OR, XOR, etc. are implemented using NMOS and PMOS transistors.
- **Microprocessors & Microcontrollers:** Billions of MOSFETs are integrated to form CPUs, GPUs, and SoCs.
- **Memory devices:** MOS transistors are used in SRAM (Static RAM), DRAM (Dynamic RAM), and Flash memory.

Why CMOS?

- Very low static power consumption.
- High noise immunity.
- High packing density (small size).

CMOS Technology Application

2. Analog Circuits

MOS transistors are also widely used in analog ICs due to their high input impedance and scalability.

- **Amplifiers:** Used in differential amplifiers, operational amplifiers (op-amps), and gain stages.
- **Voltage/current mirrors:** For biasing and reference generation.
- **Analog switches and multiplexers:** Due to their voltage-controlled resistance property.
- **Mixers and oscillators:** In RF and analog front-end circuits.

3. Power Electronics (Power ICs)

Power MOSFETs are used in integrated circuits for power management and switching.

- **Voltage regulators** (e.g., DC–DC converters).
- **Power drivers** for motors, LEDs, and loads.
- **Switching regulators** and **battery management ICs**.

CMOS Technology Application

4. Memory Circuits

MOS transistors serve as **storage and access elements** in semiconductor memories:

- **DRAM:** One MOSFET + one capacitor per bit.
- **SRAM:** Six-transistor cell (6T) built from CMOS inverters and access transistors.
- **Flash memory:** Uses floating-gate MOS transistors to store charge (non-volatile).

5. Mixed-Signal ICs

MOSFETs are used in circuits that combine analog and digital functionalities:

- **Data converters (ADC/DAC).**
- **Phase-locked loops (PLL).**
- **Signal processing circuits.**

CMOS Technology Application

6. RF and Communication ICs

In RF ICs (used in wireless communication devices):

- MOSFETs are used as **low-noise amplifiers (LNA)**, **mixers**, and **oscillators**.
- Modern RF CMOS technology integrates analog, digital, and RF circuits on a single chip.

7. Sensors and Interface Circuits

MOSFETs are also used in:

- **CMOS image sensors.**
- **Bio-sensors and chemical sensors** (where gate voltage is affected by environmental changes).

CMOS Amplifier

In general, an amplifier produces an output (voltage or current) that is a magnified version of the input (voltage or current). Since most electronic circuits both sense and produce voltage quantities, “voltage gain,” *vout/vin*.

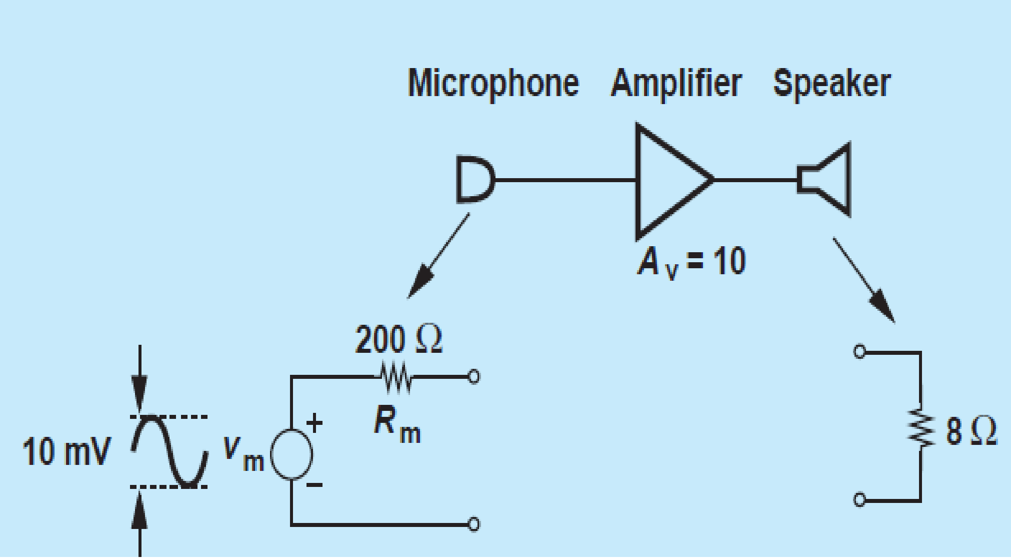
What amplifier’s performance are important?

Three parameters:

(1) power dissipation (e.g., because it determines the battery lifetime in a cellphone or a digital camera);

(2) speed (must operate at high frequencies);

(3) noise (must introduce negligible noise of its own).



CMOS Amplifier

The I/O impedances of a voltage amplifier may considerably depart from the ideal values, requiring attention to the interface with other stages.

$$v_1 = \frac{R_{in}}{R_{in} + R_m} v_m.$$

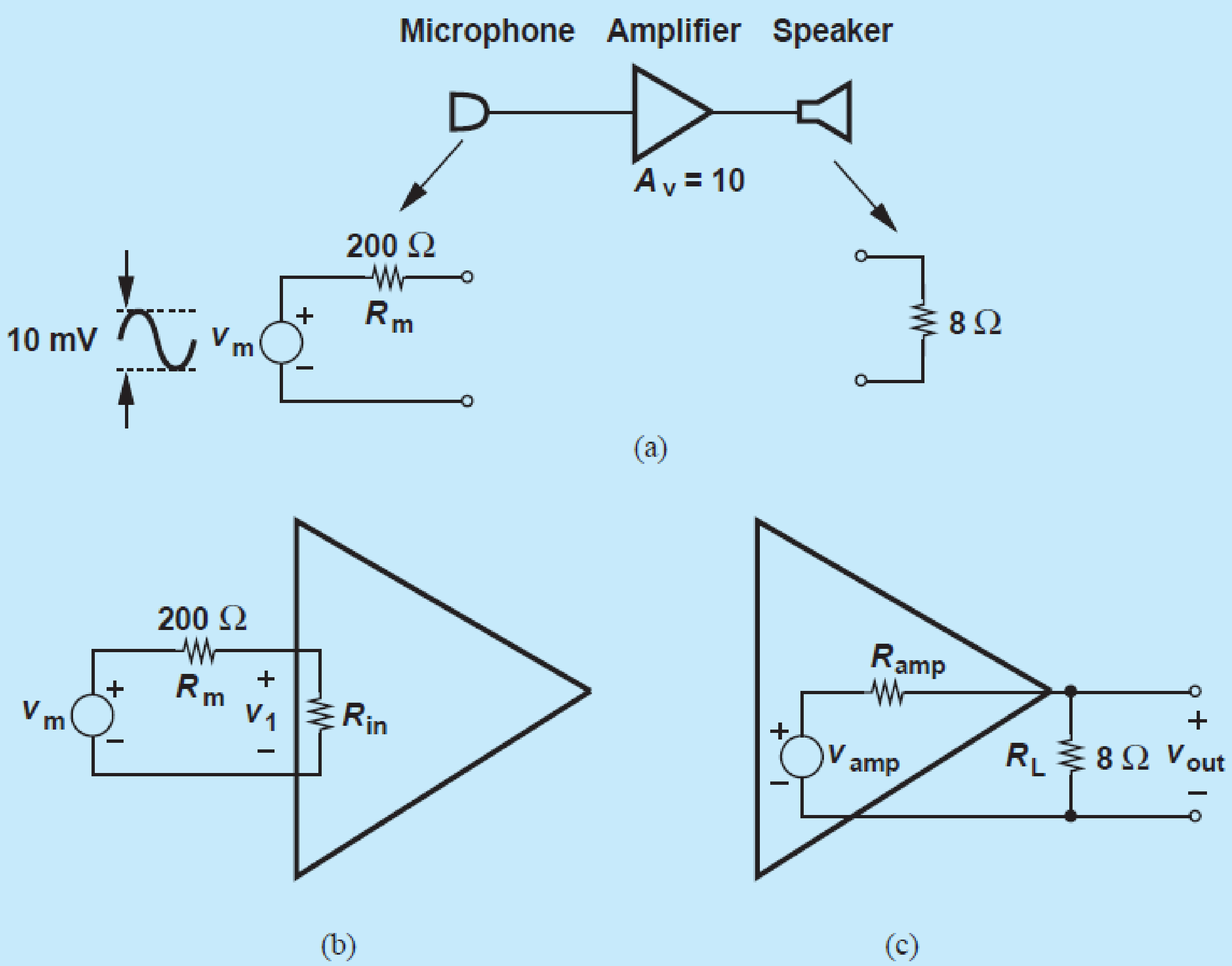
For $R_{in} = 2\,\mathrm{k}\Omega$,

$$v_1 = 0.91 v_m,$$

only 9% less than the microphone signal level.

For *Rin* =500Ω ,

$$v_1 = 0.71 v_m,$$



(a)
(b)
(c)

CMOS Amplifier

At the output;

$$v_{out} = \frac{R_L}{R_L + R_{amp}} v_{amp}.$$

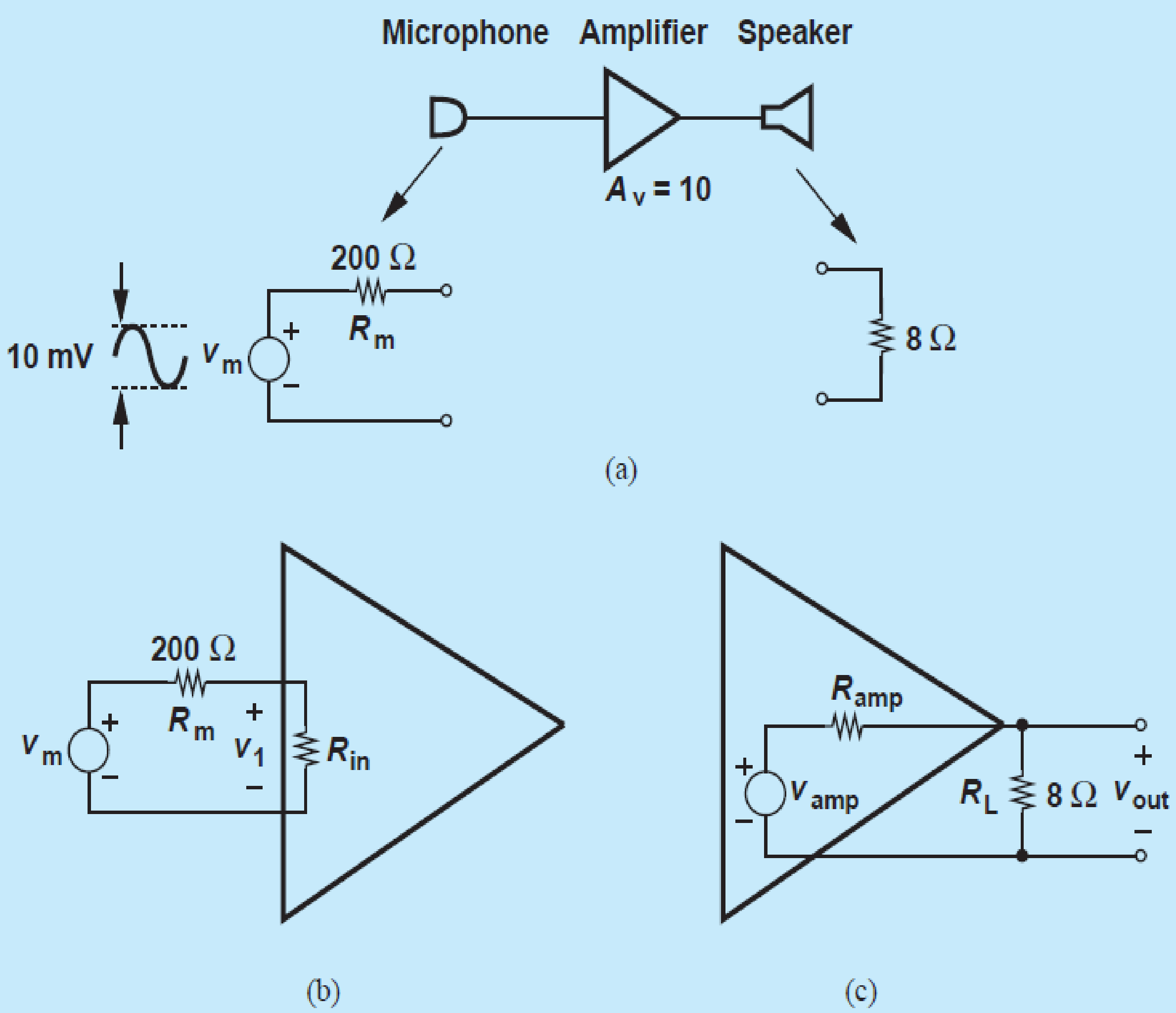
For $R_{amp} = 10\,\Omega$,

$$v_{out} = 0.44 v_{amp},$$

a substantial attenuation. For $R_{amp} = 2\,\Omega$,

$$v_{out} = 0.8 v_{amp}.$$

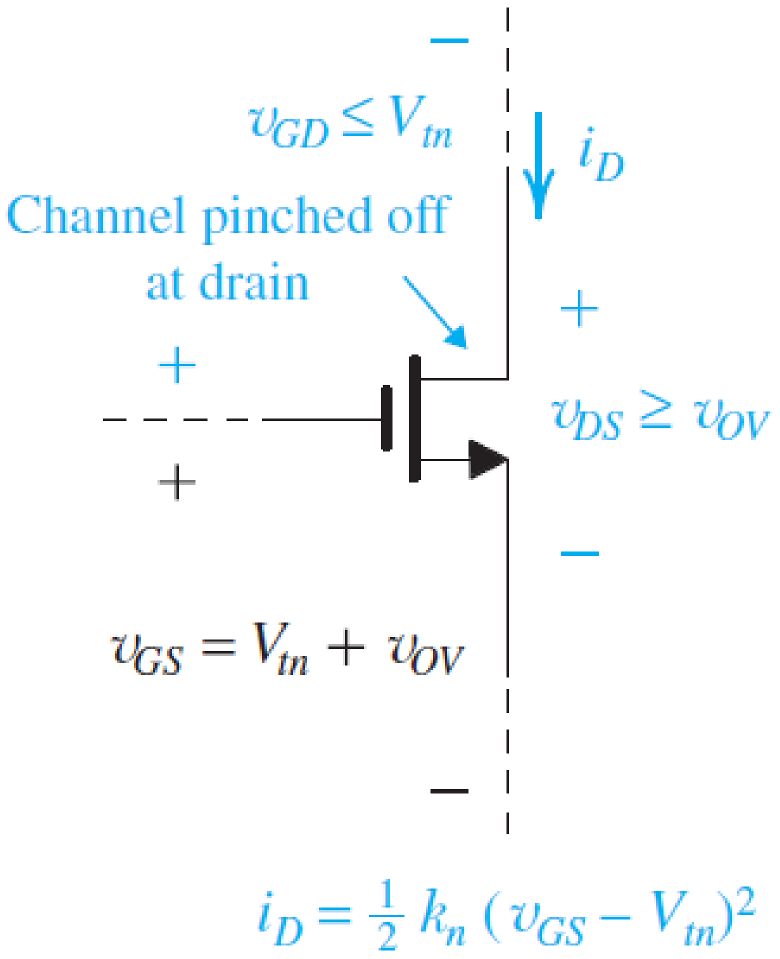
Thus, the output impedance of the amplifier must be minimized.



CMOS Amplifier

The basis for the application of the transistor MOSFET in amplifier design is that when the device is operated in the active region, a voltage-controlled current source is realized.

Specifically, when a MOSFET is operated in the saturation or pinch-off region, also referred to the active region, the voltage between gate and source, v_{GS}, controls the drain current i_D



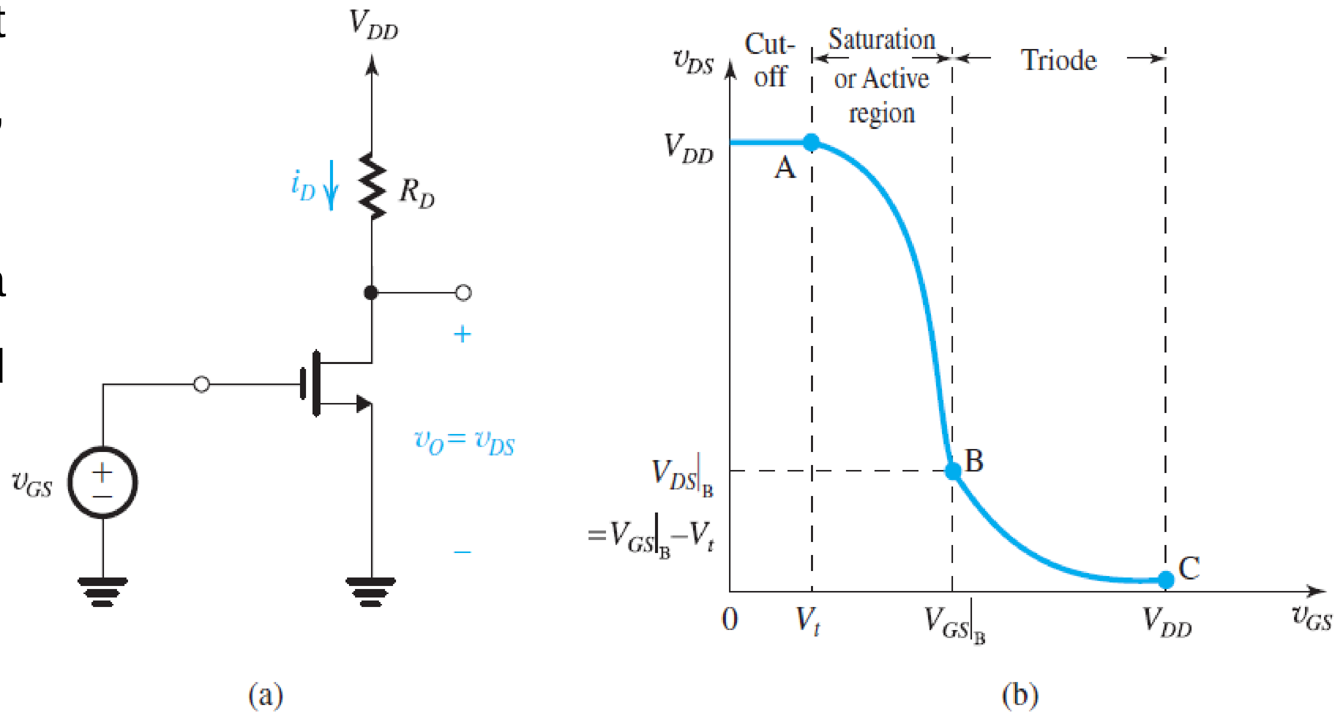
Voltage Amplifier in Technology CMOS

- In the amplifier circuit of Fig. at right the output voltage is taken between the drain and ground, rather than simply across R_D.
- This is done because of the need to maintain a common ground reference between the input and the output. The output voltage v_{DS} is given by

$$v_{DS} = V_{DD} - i_D R_D$$

- The VTC in Fig. indicates that the segment of greatest slope is that labeled AB, which corresponds to operation in the active region.

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2$$



(**a**) An NMOS amplifier, (**b**)Voltage Transfer Characteristic (VTC);

Voltage Amplifier in Technology CMOS

- With

$$v_{DS} = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_t)^2$$

$$v_{GS} = V_{GS}\big|_B \text{ and } v_{DS} = V_{DS}\big|_B = V_{GS}\big|_B - V_t$$

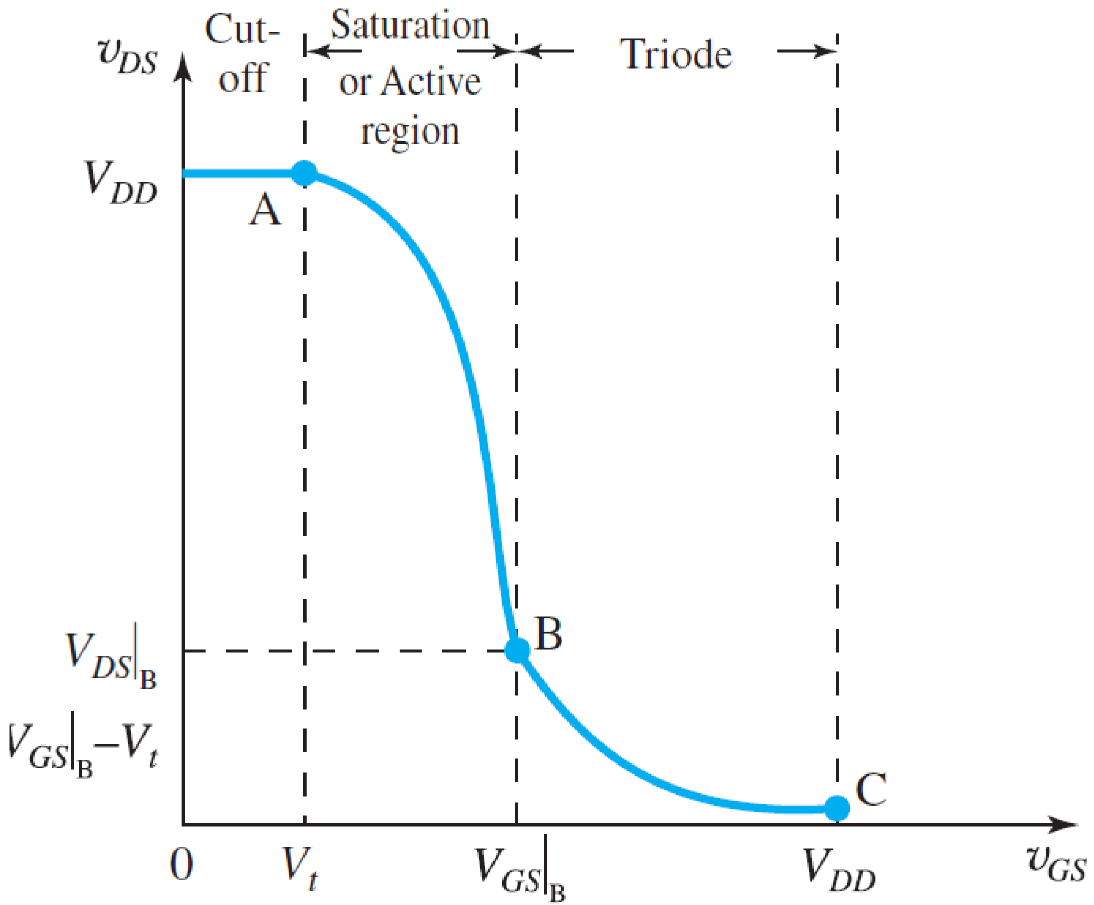
$$V_{GS}\big|_B = V_t + \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D}$$

Point B can be alternatively characterized by the overdrive voltage

$$V_{OV}\big|_B \equiv V_{GS}\big|_B - V_t = \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D}$$

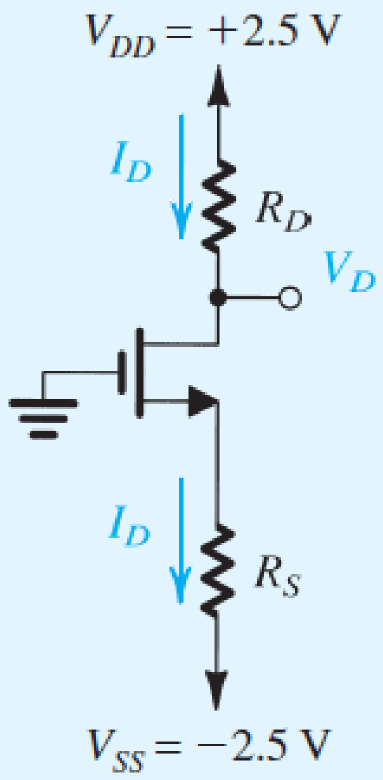
and

$$V_{DS}\big|_B = V_{OV}\big|_B$$



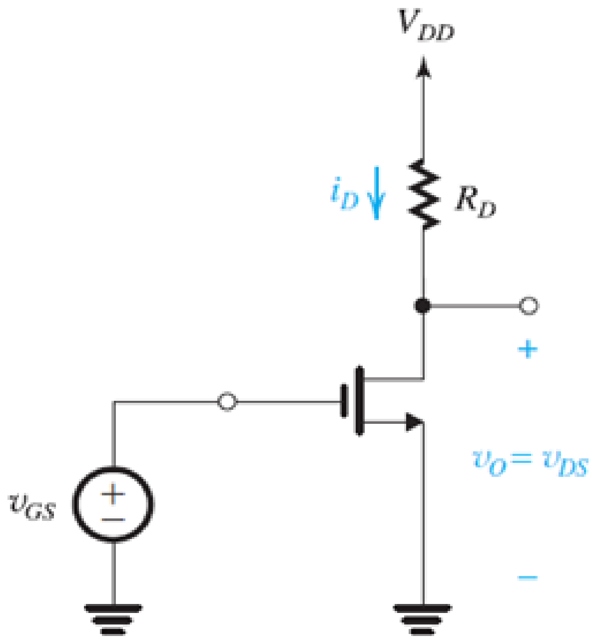
Example 5

Design the circuit of Fig.below that is, determine the values of R_D and R_S so that the transistor operates at $I_D = 0.4$ mA and $V_D = +0.5$ V. The NMOS transistor has $V_t = 0.7$ V, $\mu_n C_{ox} = 100\ \mu A/V^2$, $L = 1\ \mu m$, and $W = 32\ \mu m$. Neglect the channel-length modulation effect (i.e., assume that $\lambda = 0$).



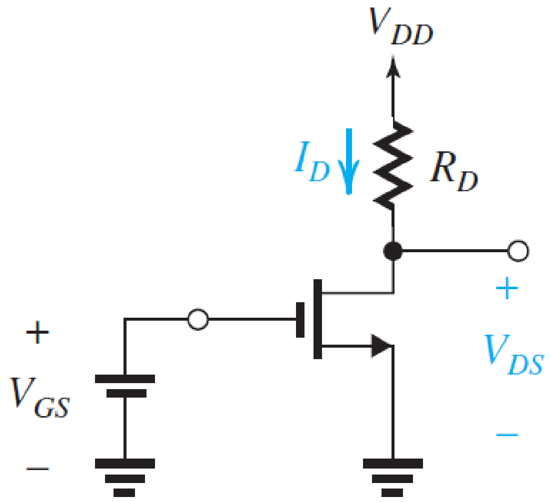
Exercise

Consider the amplifier of Fig.below with $V_{DD} = 1.8$ V, $R_D = 17.5$ kΩ, and with a MOSFET specified to have $V_t = 0.4$ V, $k_n = 4$ mA/V^2, and $\lambda = 0$. Determine the coordinates of the end points of the active-region segment of the VTC. Also, determine $V_{DS}\big|_C$ assuming $V_{GS}\big|_C = V_{DD}$.



Voltage Amplifier in Technology CMOS

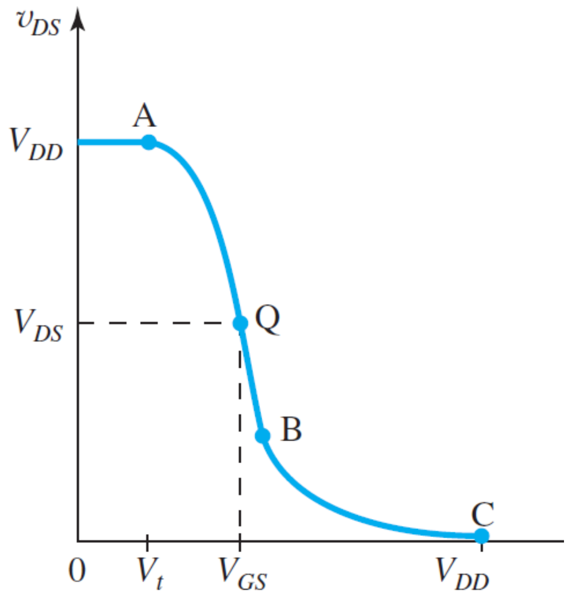
Linear Amplification by Biasing the Transistor



- Biasing enables us to obtain almost-linear amplification from the MOSFET
- A dc voltage V_{GS} is selected to obtain operation at a point Q on the segment AB of the VTC.
- Coordinates of Q are the dc voltages V_{GS} and V_{DS}, which are related by

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2$$

- Point Q is known as the **bias point** or the **dc operating point**.
- At Q no signal component is present, it is also known as the **quiescent point** (which is the origin of the symbol Q).



Voltage Amplifier in Technology CMOS

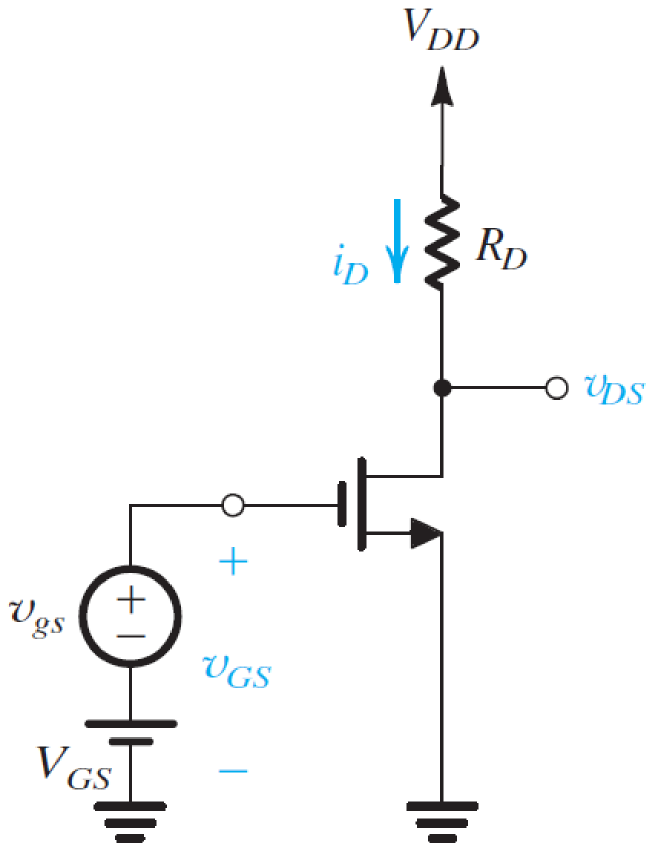
Linear Amplification by Biasing the Transistor

- The signal to be amplified, v_{gs}, a function of time t, is superimposed on the bias voltage V_{GS}, as shown in Fig. at right.
- The total instantaneous value of v_{GS} becomes

$$v_{GS}(t) = V_{GS} + v_{gs}(t)$$

- $v_{DS}(t)$ can be obtained by substituting for $v_{GS}(t)$ into Eq.

$$v_{DS} = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_t)^2$$

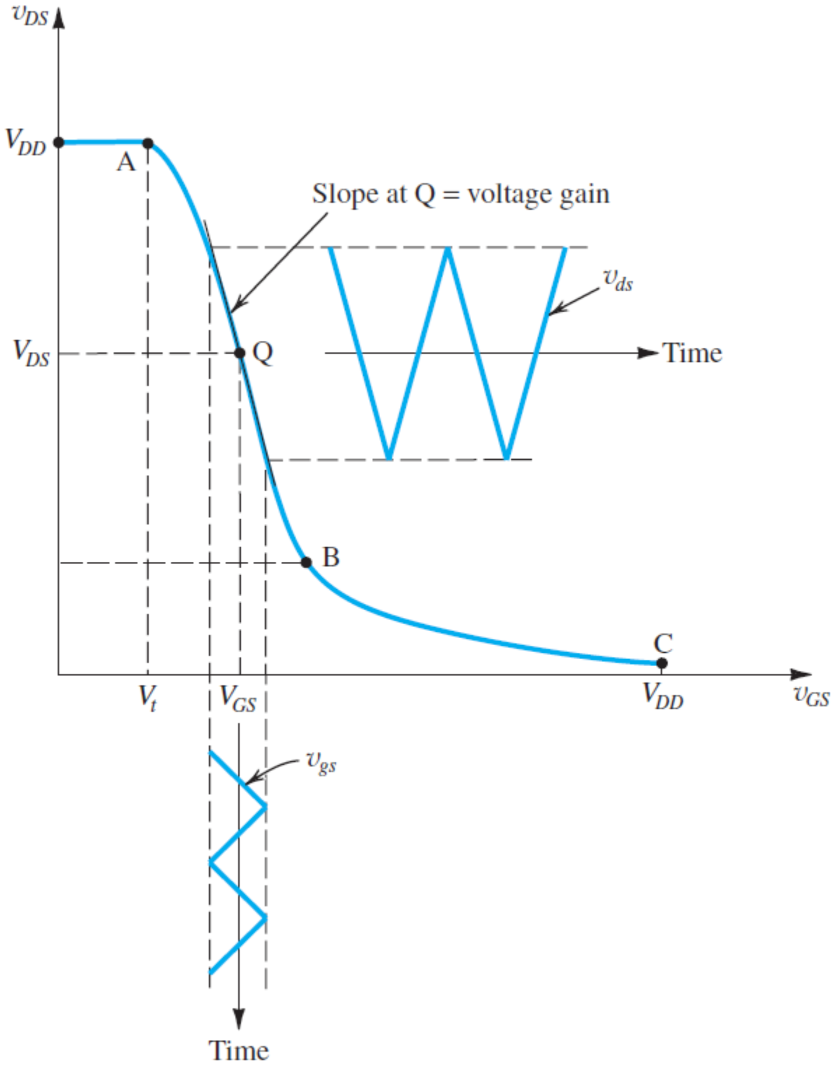


Common Source (CS) Amplifier

Voltage Amplifier in Technology CMOS

Linear Amplification by Biasing the Transistor

- Graphically, we can use the VTC to obtain $v_{DS}(t)$ point by point, as illustrated in Fig. at right
- The MOSFET amplifier with a small time-varying signal $v_{gs}(t)$ superimposed on the dc bias voltage V_{GS}. The MOSFET operates on a short almost-linear segment of the VTC around the bias point Q and provides an output voltage $v_{ds} = Av*v_{gs}$.



Voltage Amplifier in Technology CMOS

Small-Signal Voltage Gain

- Consider the MOSFET amplifier (CS),
  - If the input signal vgs is kept small, the corresponding signal at the output v_{ds} will be nearly proportional to v_{gs} with the constant of proportionality being the slope of the almost-linear segment of the VTC around Q.
  - voltage gain of the amplifier, and its value can be determined by evaluating the slope of the tangent to the VTC at the bias point Q

$$A_v = \left.\frac{dv_{DS}}{dv_{GS}}\right|_{v_{GS}=V_{GS}}$$

- With $$v_{DS} = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_t)^2$$

$$A_v = -k_n(V_{GS} - V_t)R_D = -k_n V_{OV} R_D$$

With $$I_D = \frac{1}{2}k_n V_{OV}^2$$

Small Signal Voltage gain is given by

$$A_v = -\frac{I_D R_D}{V_{OV}/2}$$

Voltage Amplifier in Technology CMOS

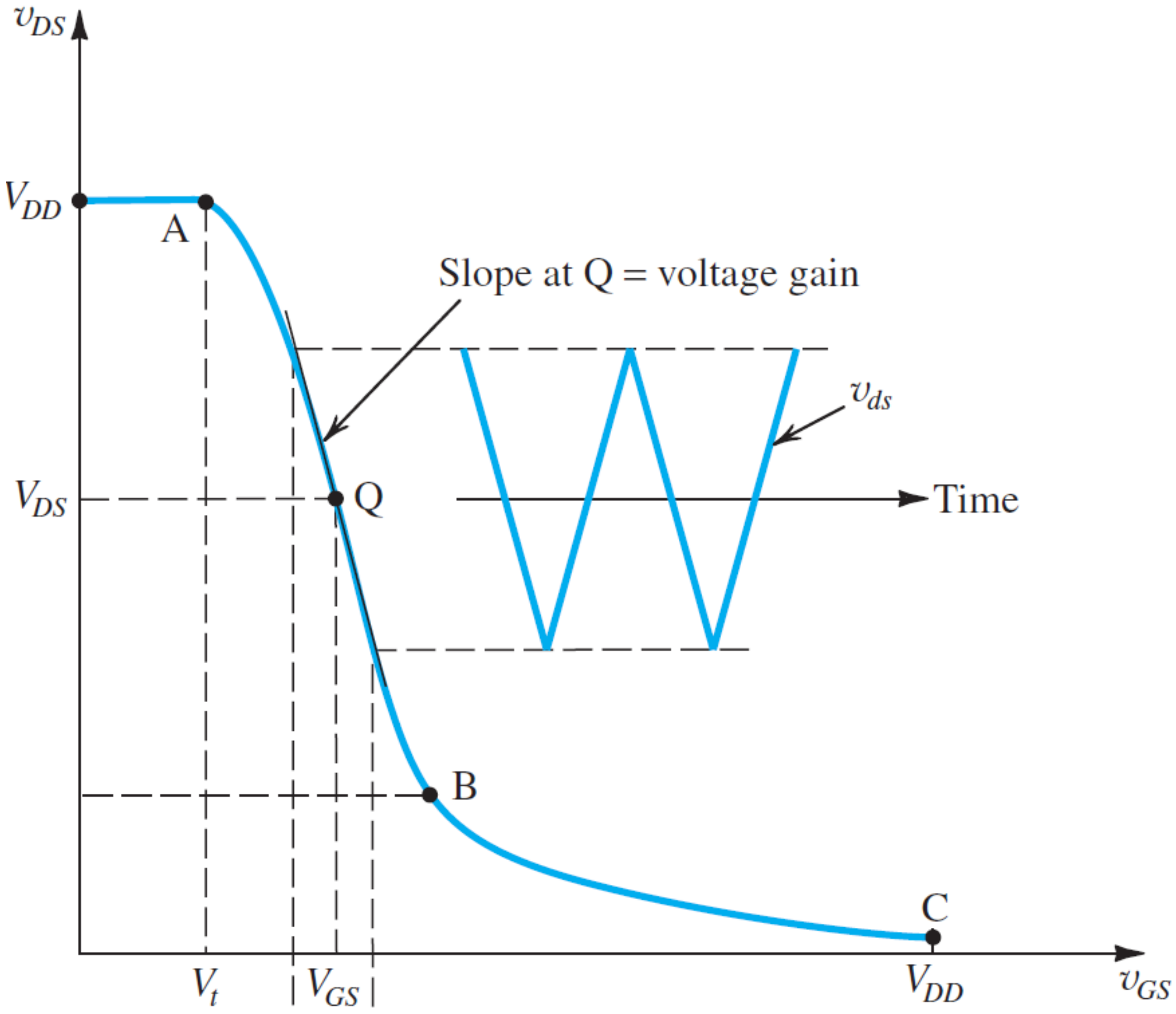
Small-Signal Voltage Gain

- That is, the gain is simply the ratio of the dc voltage drop across the load resistance RD to $V_{OV}/2$. It can be expressed in the alternative form

$$A_v = -\frac{V_{DD} - V_{DS}}{V_{OV}/2}$$

- Since the maximum slope of the VTC in Fig. occurs at point B, the maximum gain magnitude $|A_{vmax}|$ is obtained by biasing the transistor at point B,

$$|A_{v\max}| = \frac{V_{DD} - V_{DS}|_B}{V_{OV}|_B/2}$$



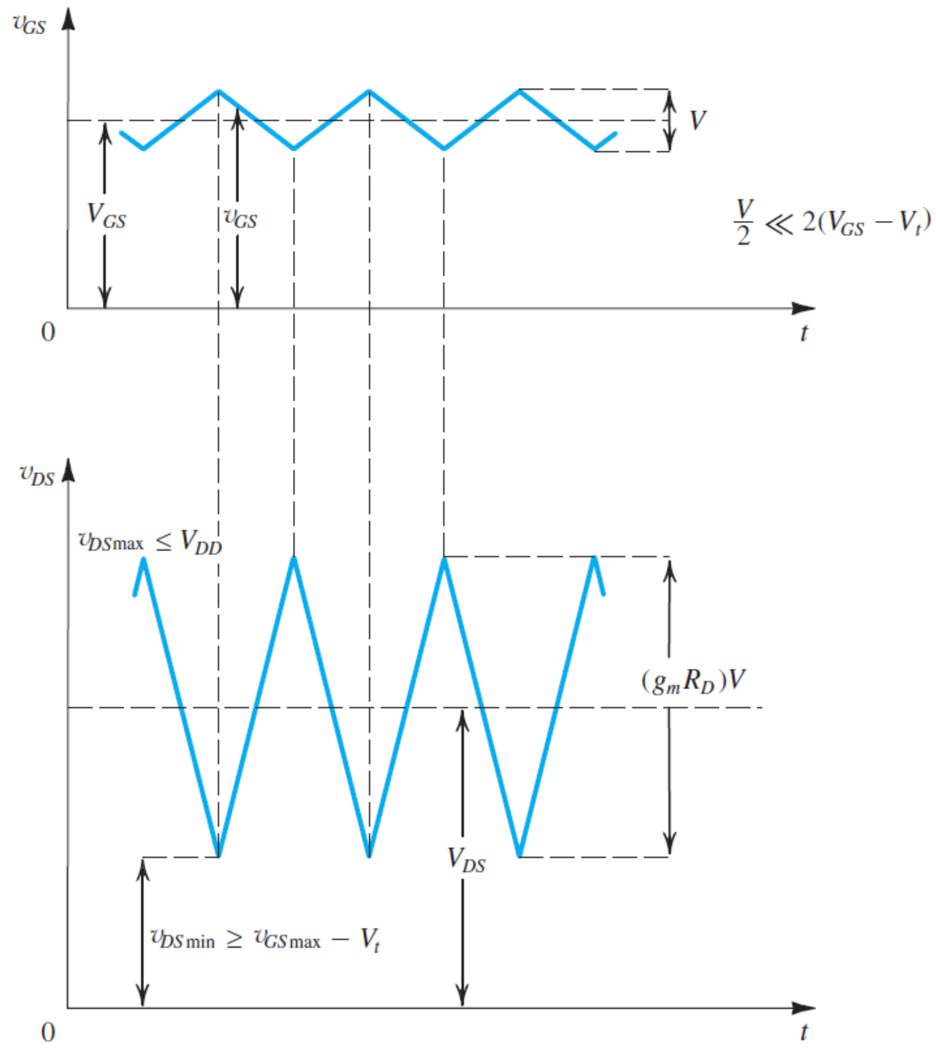
Voltage Amplifier in Technology CMOS

Small-Signal Voltage Gain

$$A_v \equiv \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

Example :

- The input signal is assumed to have a triangular waveform with an amplitude much smaller than $2(V_{GS} - V_t)$, to ensure linear operation.
- For operation in the saturation (active) region at all times, the minimum value of v_{DS} should not fall below the corresponding value of v_{GS} by more than V_t.
- Also, the maximum value of v_{DS} should be smaller than *VDD*



Voltage Amplifier in Technology CMOS

Example 6

Consider the amplifier circuit shown in Fig. 7.4(a). The transistor is specified to have $V_t = 0.4$ V, $k_n' = 0.4$ mA/V^2, $W/L = 10$, and $\lambda = 0$. Also, let $V_{DD} = 1.8$ V, $R_D = 17.5$ kΩ, and $V_{GS} = 0.6$ V.

(a) For $v_{gs} = 0$ (and hence $v_{ds} = 0$), find V_{OV}, I_D, V_{DS}, and A_v.

(b) What is the maximum symmetrical signal swing allowed at the drain?

Voltage Amplifier in Technology CMOS

Example 6

Solution

(a) With $V_{GS} = 0.6$ V, $V_{OV} = 0.6 - 0.4 = 0.2$ V. Thus,

$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) V_{OV}^2$$
$$= \frac{1}{2} \times 0.4 \times 10 \times 0.2^2 = 0.08 \text{ mA}$$
$$V_{DS} = V_{DD} - R_D I_D$$
$$= 1.8 - 17.5 \times 0.08 = 0.4 \text{ V}$$

Since V_{DS} is greater than V_{OV}, the transistor is indeed operating in saturation. The voltage gain can be found from Eq.

$$A_v = -k_n V_{OV} R_D$$
$$= -0.4 \times 10 \times 0.2 \times 17.5$$
$$= -14 \text{ V/V}$$

Small-Signal Operation and Models

Signal Current in the Drain Terminal

- The total instantaneous gate-to-source voltage will be $v_{GS} = V_{GS} + v_{gs}$
- resulting in a total instantaneous drain current i_D,

$$i_D = \frac{1}{2}k_n\left(V_{GS} + v_{gs} - V_t\right)^2$$

$$= \frac{1}{2}k_n(V_{GS} - V_t)^2 + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2$$

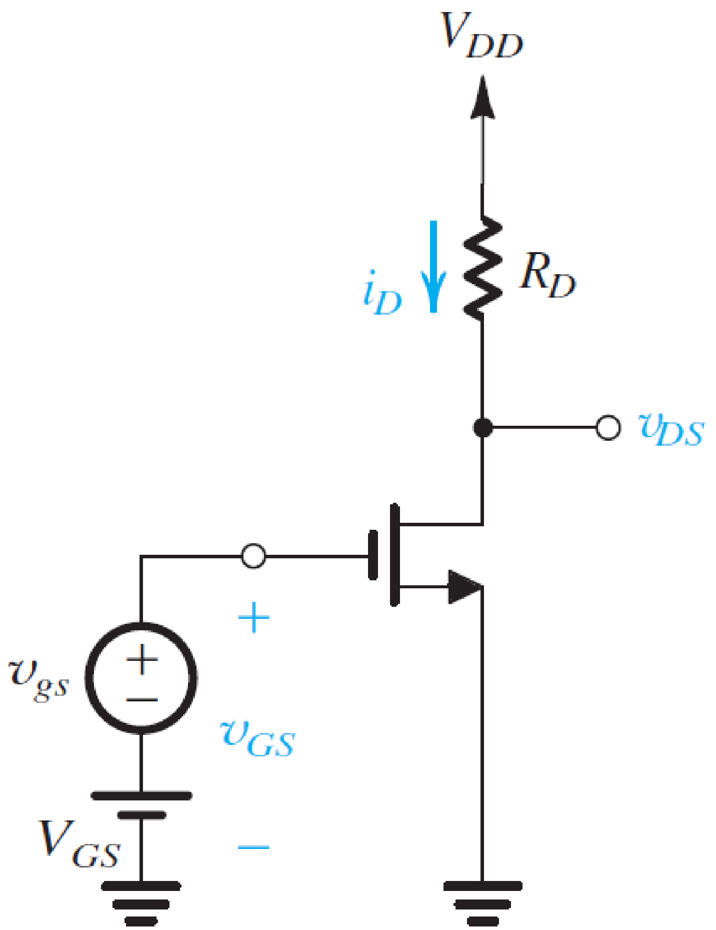
With $v_{gs} \ll 2V_{OV}$

With the small-signal condition is satisfied, i_D is expressed as

$$i_D \simeq I_D + i_d$$

where

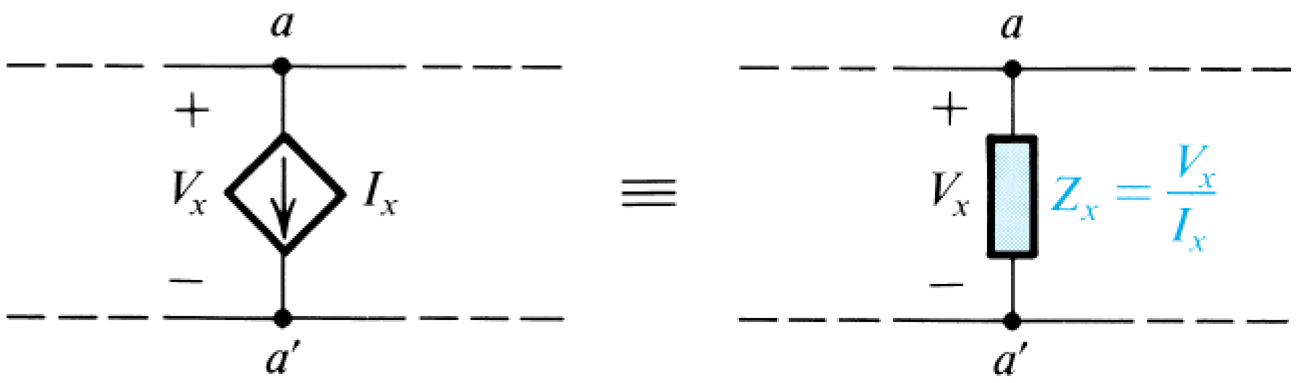
$$i_d = k_n(V_{GS} - V_t)v_{gs}$$



Small-Signal Operation and Models

The T Equivalent-Circuit Model

Source-Absorption Theorem/ Appendix D.3 (Microelectronic Circuits, seventh edition)



I_x appearing between two nodes whose voltage difference is the controlling voltage V_x. That is,

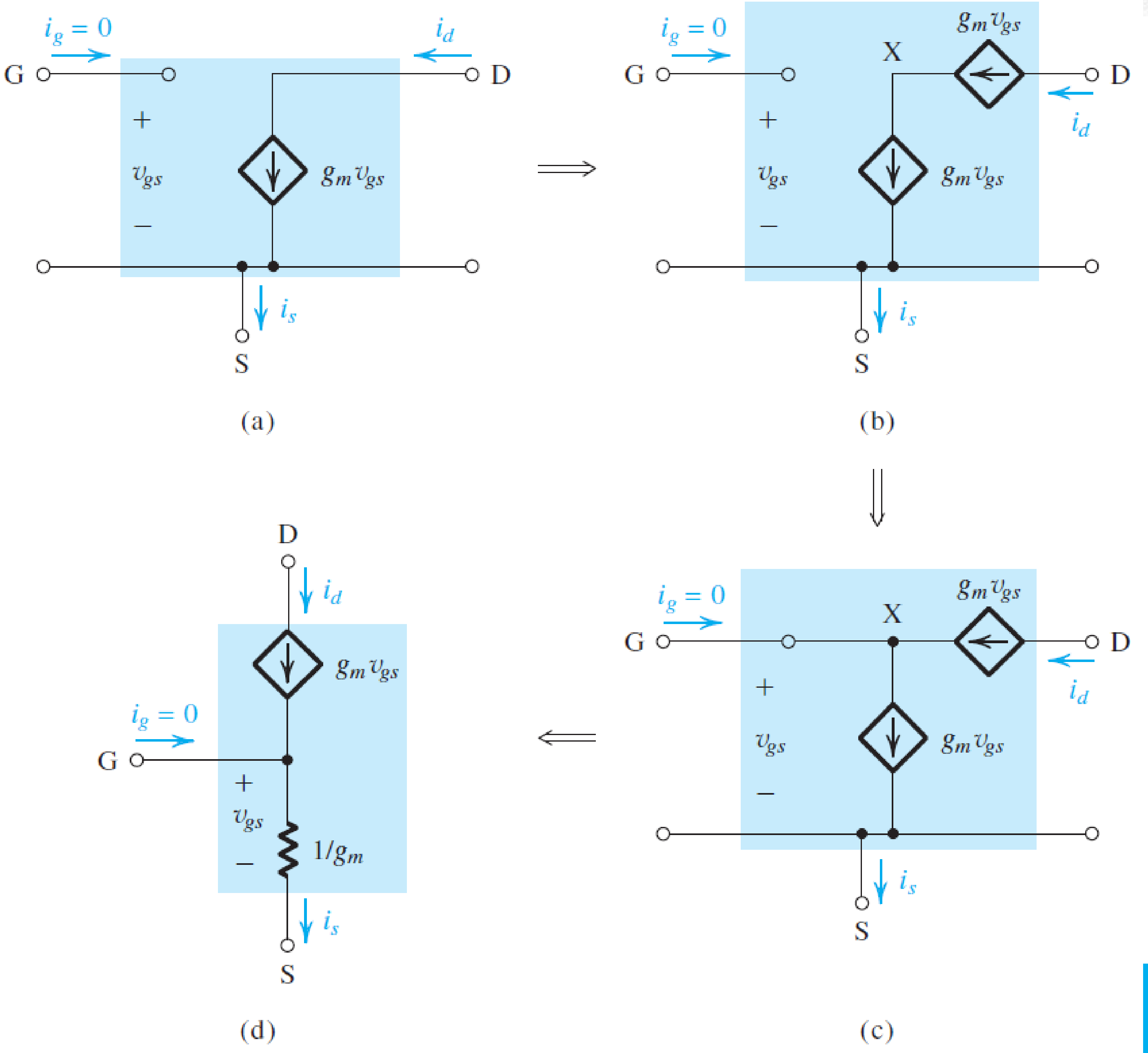
$I_x = g_m V_x$ where g_m is a conductance. We can replace this controlled source by an impedance

$Z_x = V_x/I_x = 1/g_m$,

Small-Signal Operation and Models

The T Equivalent-Circuit Model

- Development of the T equivalent-circuit model for the MOSFET. For simplicity, *ro* has been omitted; however, it may be added between D and S in the T model of **(d)**.



Small-Signal Operation and Models

Transconductance *gm*

The parameter that relates i_d and v_{gs} is the MOSFET **transconductance** g_m,

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n(V_{GS} - V_t)$$

$$g_m = k_n'(W/L)(V_{GS} - V_t) = k_n'(W/L)V_{OV}$$

$$g_m = \sqrt{2I_D/(k_n'(W/L))} = \sqrt{2k_n'}\sqrt{W/L}\sqrt{I_D} = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}}$$

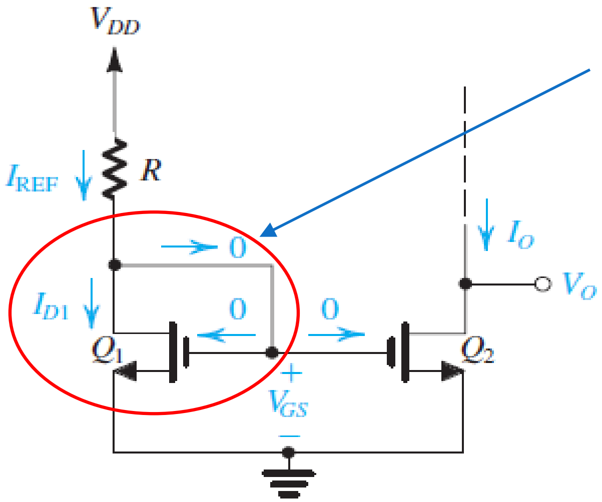
This expression shows two things:

1. For a given MOSFET, g_m is proportional to the square root of the dc bias current.
2. At a given bias current, g_m is proportional to $\sqrt{W/L}$.

IC Biasing—Current Sources, Current Mirrors

The Basic MOSFET Current Source

MOSFET diode connected



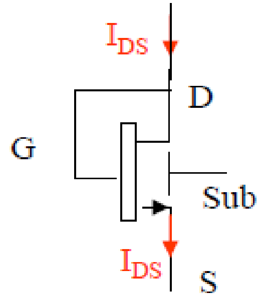
Circuit for a basic MOSFET constant current source. For proper operation, the output terminal, that is, the drain of $Q2$, must be connected to a circuit that ensures that $Q2$ operates in saturation (.

$$I_{D1} = \frac{1}{2} k_n' \left(\frac{W}{L}\right)_1 (V_{GS} - V_{tn})^2$$

With λ=0 ; $V_O \geq V_{GS} - V_{tn}$

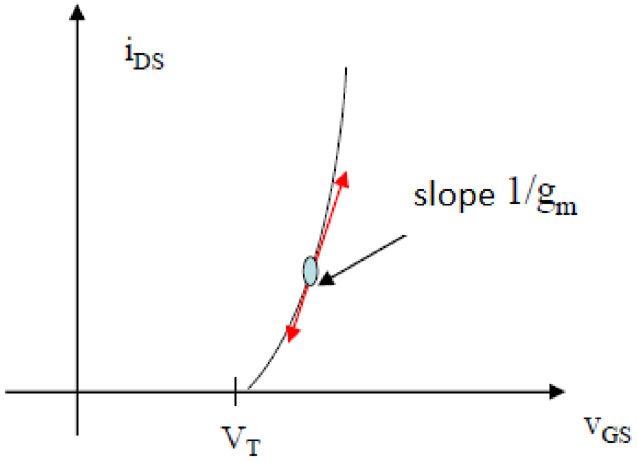
$$I_{D1} = I_{REF} = \frac{V_{DD} - V_{GS}}{R} \rightarrow \frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

MOSFET diode connected



$$v_{GS} = v_{DS}$$

$$i_{DS} = \frac{W}{2L} \mu C_{OX} (v_{GS} - V_T)^2$$

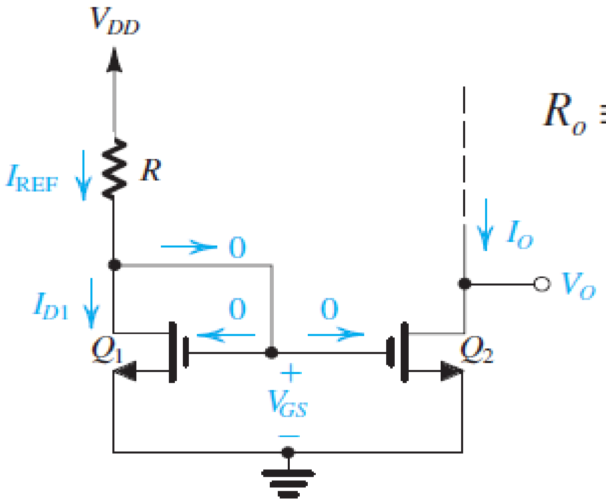


$$r = \frac{dv_{GS}}{di_{DS}} = \frac{1}{g_m};$$ dynamic resistance

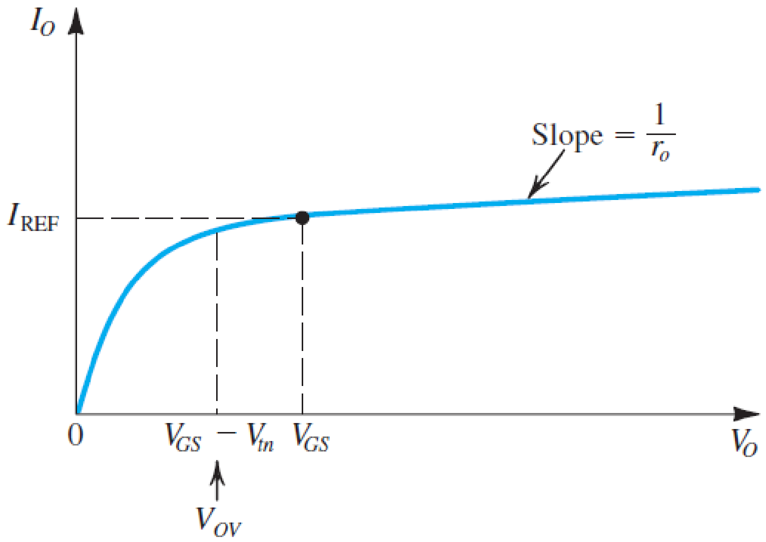
$$r = \frac{L}{W} \frac{1}{\mu C_{OX} (V_{GS} - V_T)}$$

IC Biasing—Current Sources, Current Mirrors

The Basic MOSFET Current Source



$$R_o \equiv \frac{\Delta V_O}{\Delta I_O} = r_{o2} = \frac{V_{A2}}{I_O}$$



the current source have a finite output resistance R_o

Output characteristic of the current source for the case of Q_2 matched to Q_1.

- Consider, for simplicity, the case of identical devices Q_1 and Q_2.
- The drain current of Q_2, I_O, will equal the current in Q_1, I_{REF}, at the value of V_O that causes the two devices to have the same V_{DS}, that is, at $V_O = V_{GS}$.
- As V_O is increased above this value, I_O will increase according to the incremental output resistance r_{o2} of Q_2

IC Biasing—Current Sources, Current Mirrors

Example 7

Given $V_{DD} = 3$ V and using $I_{REF} = 100\ \mu A$, design the circuit of Fig.1 to obtain an output current whose nominal value is 100 μA. Find R if Q_1 and Q_2 are matched and have channel lengths of 1 μm, channel widths of 10 μm, $V_t = 0.7$ V, and $k'_n = 200\ \mu A/V^2$. What is the lowest possible value of V_O? Assuming that for this process technology, the Early voltage $V'_A = 20$ V/μm, find the output resistance of the current source. Also, find the change in output current resulting from a +1-V change in V_O.

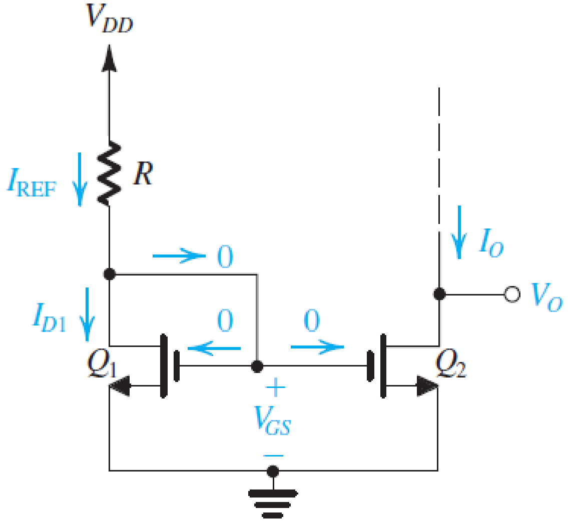


Fig 1

IC Biasing—Current Sources, Current Mirrors

example 7

Solution

$$I_{D1} = I_{REF} = \frac{1}{2} k_n' \left(\frac{W}{L}\right)_1 V_{OV}^2$$

$$100 = \frac{1}{2} \times 200 \times 10\, V_{OV}^2$$

Thus,

$$V_{OV} = 0.316 \text{ V}$$

and

$$V_{GS} = V_t + V_{OV} = 0.7 + .316 \simeq 1 \text{ V}$$

$$R = \frac{V_{DD} - V_{GS}}{I_{REF}} = \frac{3 - 1}{0.1 \text{ mA}} = 20 \text{ k}\Omega$$

$$V_{O\min} = V_{OV} \simeq 0.3 \text{ V}$$

For the transistors used, $L = 1\ \mu\text{m}$. Thus,

$$V_A = 20 \times 1 = 20 \text{ V}$$

$$r_{o2} = \frac{20 \text{ V}}{100\ \mu\text{A}} = 0.2 \text{ M}\Omega$$

The output current will be $100\ \mu\text{A}$ at $V_O = V_{GS} = 1$ V. If V_O changes by $+1$ V, the corresponding change in I_O will be

$$\Delta I_O = \frac{\Delta V_O}{r_{o2}} = \frac{1 \text{ V}}{0.2 \text{ M}\Omega} = 5\ \mu\text{A}$$